2025/09 PHY301 Thermodynamic and Statistical Physics

Problems set 2

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Problem 1: Classical harmonic oscillator in Canonical ensemble [25 points]

Consider N harmonic oscillators with coordinates and momenta $\{q_i, p_i\}$ and subject to a Hamiltonian

$$\mathcal{H}(\{q_i, p_i\}) = \sum_{i=1}^{N} \left[\frac{p_i^2}{2m} + \frac{m\omega^2 q_i^2}{2} \right]$$

(a) In canonical ensemble, calculate the partition function $\mathcal{Z}$ for the classical osicillator using

$$\mathcal{Z} = \int \frac{d^{3N}q d^{3N}p}{h^3} \mathrm{e}^{-\beta\mathcal{H}}$$

Similar to the previous assignment, the integration is easier to carry out after and appropiate transformation.

Solution

$$\begin{aligned}\mathcal{Z} =& \prod_{i=1}^{N} \int \frac{d^3p_i d^3q_i}{h^3} \exp\Big[-\beta\Big(\frac{p_i^2}{2m} + m\omega^2 \frac{q_i^2}{2} \Big)\Big] \\ =& \prod_{i=1}^{N} \Big(\frac{\pi}{\beta/(2m)} \frac{\pi}{\beta m\omega^2/2} \frac{1}{h^2} \Big)^{3/2}\Big] = \Big(\frac{2\pi}{\beta\hbar\omega} \Big)^{3N}\end{aligned}$$

[7 points]

(b) Using (a), calculate the Helmholtz free energy F, as a function of total energy E.

Solution

$$F = -k_B T \ln \mathcal{Z} = -3Nk_B T \ln\Big(\frac{1}{\beta\hbar\omega} \Big)$$

The total energy is

$$E = -\frac{d\ln\mathcal{Z}}{d\beta} = 3N \frac{d}{d\beta} \ln \beta\hbar\omega = 3Nk_B T$$

We then

$$F = -E \ln\Big(\frac{E}{3N\hbar\omega} \Big)$$

[6 points]

(c) Calculate the entropy S, and heat capacity C.

Solution

The entropy is

$$S = -\frac{\partial F}{\partial T} = 3Nk_B \ln \frac{k_B T}{\hbar\omega} + 3Nk_B$$

Heat capacity is

$$C_T = \frac{dE}{dT} = 3Nk_B$$

[6 points]

(d) Find the joint probability density $P(p, q)$ for a single oscillator. Hence calculate the mean kinetic energy, and mean potential energy for each oscillator.

Solution

The joined probability is

$$p(p,q) = \frac{1}{\mathcal{Z}} \prod_{j\neq i} \int \frac{d^3p_j d^3q_j}{h^3} \exp\left[-\beta\left(\frac{p_j^2}{2m} + m\omega^2 \frac{q_j^2}{2}\right)\right] = \frac{(\hbar\omega)^3}{(k_B T)^3} \exp\left[-\beta\left(\frac{p_j^2}{2m} + m\omega^2 \frac{q_j^2}{2}\right)\right]$$

[6 points]

Problem 2: Quantum harmonic oscillators in Canonical ensemble [25 points]

Consider N independent quantum oscillators subjects to a Hamiltonian

$$\mathcal{H}(\{n_i\}) = \sum_{i=1}^{N} \hbar\omega \left(n_i + \frac{1}{2}\right),$$

where $n_i = 0, 1, 2, \ldots$ is quantum occupation number for the ith oscillator.

(a) Using canonical ensemble, calculate the partition function $\mathcal{Z}$, as a function of temperature T.

Solution

The partition function

$$\mathcal{Z} = \sum_{n_1,\ldots n_N} \exp\Big[-\beta \sum_{i=1}^{N} \hbar\omega(n_i + \tfrac{1}{2})\Big] = \sum_{n_1,\ldots n_N} \prod_{i=1}^{N} \exp\Big[-\beta\hbar\omega(n_i + \tfrac{1}{2})\Big]$$
$$= \prod_{i=1}^{N} \sum_{n_1=0}^{\infty} \exp\Big[-\beta\hbar\omega(n_i + \tfrac{1}{2})\Big] = \prod_{i=1}^{N} \frac{\mathrm{e}^{-\frac{1}{2}\beta\hbar\omega}}{1 - \mathrm{e}^{-\beta\hbar\omega}} \quad \textbf{[8]}$$

(b) Calculate the energy E, and the heat capacity C, as functions of temperature T, and N.

Solution

The total energy

$$E = -\frac{d\ln\mathcal{Z}}{d\beta} = -N\frac{d}{d\beta}\ln\frac{e^{\frac{1}{2}\beta\hbar\omega}}{e^{\beta\hbar\omega}-1} = N\hbar\omega\Big[\frac{e^{\beta\hbar\omega}}{e^{\beta\hbar\omega}-1}-\frac{1}{2}\Big]$$

The heat capacity is

$$\begin{aligned} C_T = \frac{dE}{dT} =& -N\hbar\omega\frac{d}{dT}\Big[\frac{1}{2}-\frac{e^{\beta\hbar\omega}}{e^{\beta\hbar\omega}-1}\Big] = N\hbar\omega\frac{d}{dT}\frac{e^{\beta\hbar\omega}}{e^{\beta\hbar\omega}-1} \\ =& N\hbar\omega\Big[-\frac{1}{T}\frac{\beta\hbar\omega e^{\beta\hbar\omega}}{e^{\beta\hbar\omega}-1}+\frac{1}{T}\frac{\beta\hbar\omega e^{2\beta\hbar\omega}}{(e^{\beta\hbar\omega}-1)^2}\Big] \\ =& Nk_B\Big(\frac{\hbar\omega}{k_BT}\Big)^2\Big[\frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega}-1)^2}\Big] \quad [8] \end{aligned}$$

(c) Find the probability $p(n)$ that a particular oscillator is in its nth quantum level.

Solution

The probability

$$\begin{aligned} p(n_i) =& \frac{1}{\mathcal{Z}}\sum_{n_1}\cdots\sum_{n_{i-1}}\sum_{n_{i+1}}\cdots\sum_{n_N}\exp\Big[-\beta\sum_{i=1}^{N}\hbar\omega(n_i+\tfrac{1}{2})\Big] \\ =& \frac{1}{Z}\exp\Big[-\beta\hbar\omega(n_i+\tfrac{1}{2})\Big]\prod_{j=1,j\neq i}^{N}\sum_{n_j0}^{\infty}\exp\Big[-\beta\hbar\omega(n_i+\tfrac{1}{2})\Big] \\ =& \Big(\frac{e^{-\frac{1}{2}\beta\hbar\omega}}{1-e^{-\beta\hbar\omega}}\Big)^{-N}\exp\Big[-\beta\hbar\omega(n_i+\tfrac{1}{2})\Big]\Big(\frac{e^{-\frac{1}{2}\beta\hbar\omega}}{1-e^{-\beta\hbar\omega}}\Big)^{N-1} \\ =& e^{-\beta\hbar\omega n_i}(1-e^{-\beta\hbar\omega}) \quad [9] \end{aligned}$$

Problem 3: Grand Canonical ensemble for relativistic gas [25 points]

Gas particles can adsorb on the surface of a solid at N possible adsorption sites. Each site has binding energy ϵ, and can accommodate at most one particle. The adsorbed particles are in equilibrium with gas surrounding the solid. We can treat the system of adsorbed particles in a grand canonical ensemble with temperature T and chemical potential μ.

(a) If there are M adsorbed particles, what is the energy $E(M)$ of the system? What is the degeneracy $\Gamma(M)$ of the energy?

Solution

The total energy for M sites being occupied by the molecule is $E(M) = -M\epsilon$. The degeneracy of the this energy is a combinatorial problem of N choose M. This leads to

$$\Gamma(E) = \binom{N}{M} = \frac{N!}{M!(N-M)!} \quad \textbf{[6 points]}$$

(b) Write down the grand partition function of the system as a sum over M. Determine the thermal average $\bar{M}$ as the value that maximized the summand in the grand partition function.

Solution

The partition function for M particle system is

$$\mathcal{Z}_M = \Gamma(M)\mathrm{e}^{-\beta M\epsilon}$$

Therefore, the Grand partition function is

$$\mathcal{Q} = \sum_{M=0}^{N} \mathrm{e}^{\beta\mu M}\mathcal{Z}_M = \sum_{M=0}^{N} \binom{N}{M}\mathrm{e}^{-\beta M(\epsilon-\mu)} = (1+\mathrm{e}^{-\beta(\epsilon-\mu)})^N$$

The value M^* that maximizes $\mathrm{e}^{\beta\mu M}\mathcal{Z}_M$ can be obtained by optimizing the following

$$\frac{dg(M)}{dM} = \frac{d}{dM}\ln\left[\binom{N}{M}\mathrm{e}^{-\beta M(\epsilon-\mu)}\right]$$

where $g(M) = \ln\left[\binom{N}{M}\mathrm{e}^{-\beta M(\epsilon-\mu)}\right]$ and $\left.\frac{dg(M)}{dM}\right|_{M=M^*} = 0$. This gives

$$\begin{aligned}\frac{dg(M)}{dM} \approx& \frac{d}{dM}\Big[N\ln N - (N-M)\ln(N-M) - M\ln M - \beta M(\epsilon-\mu)\Big]\\ \frac{dg(M)}{dM} \approx& \ln(N-M) - \ln M - \beta(\epsilon-\mu)\Big]\end{aligned}$$

Solution

Therefore, we have

$$\begin{aligned}\ln(N/M^* - 1) - \beta(\epsilon - \mu) =& 0 \\ N/M^* - 1 =& e^{\beta(\epsilon-\mu)} \\ M^* =& \frac{N}{e^{\beta(\epsilon-\mu)} + 1}\end{aligned}$$

For the average $\bar{M}$, we can calculate from

$$\bar{M} = \frac{d \ln \mathcal{Q}}{\beta d\mu} = N \frac{d \ln\left(1 + e^{-\beta(\epsilon-\mu)}\right)}{\beta d\mu} = \frac{N}{e^{\beta(\epsilon-\mu)} + 1}$$

[7 points]

(c) Suppose the massless relativistic gas with Hamiltonian $\mathcal{H} = \sum_{i=0}^{N} p_i c$ surrounding the solid has pressure P. Calculate $\bar{M}$ using the ideal-gas expression for μ.

Solution

In Canonical ensemble, the ideal gas partition function is

$$\mathcal{Z}_N = \int d^{3N} p d^{3N} q \frac{1}{h^3 N!} \exp\left[-\beta \sum_{i=1}^{N} p_i c \right] = \frac{1}{N!} \left[\frac{8\pi V}{(h\beta c)^3} \right]^N$$

In Grand Canonical ensemble, an ideal gas have the grand partition function as

$$\mathcal{Q}_{gas} = \sum_{N=0}^{\infty} e^{\beta\mu N} \mathcal{Z}_N = \exp\left[8\pi e^{\beta\mu} V/(h\beta c)^3 \right]$$

where $\lambda = h/\sqrt{2\pi m k_B T}$. The Grand potential is

$$\mathcal{G} = -k_B T \ln \mathcal{Q}_{gas} = -\frac{1}{\beta} e^{\beta\mu_g} V/\lambda^3$$

On the other hand, we have

$$\mathcal{G} = -PV$$

This gives the chemical potential of the gas as

$$\begin{aligned}P =& \frac{1}{\beta\lambda^3} e^{\beta\mu_g} \\ \mu_g =& \frac{1}{\beta} \ln \beta P \lambda^3\end{aligned}$$

Solution

In thermal equilibrium, $\mu_g = \mu$. Therefore, we have

$$\begin{aligned}\bar{M} =& \frac{N}{\mathrm{e}^{\beta(\epsilon-\mu_g)}+1} \\ =& \frac{N}{\mathrm{e}^{-\beta\mu_g}\mathrm{e}^{\beta\epsilon}+1} \\ =& \frac{N\beta P\lambda^3}{\mathrm{e}^{\beta\epsilon}+\beta P\lambda^3}\end{aligned}$$

[7 points]

(d) Find $\overline{M^2} - \overline{M}^2$.

Solution

$$\overline{M^2} - \overline{M}^2 = \frac{d\overline{M}}{\beta d\mu} = \frac{d}{\beta d\mu}\frac{N}{\mathrm{e}^{\beta(\epsilon-\mu)}+1} = -\frac{-N\mathrm{e}^{\beta(\epsilon-\mu)}}{(\mathrm{e}^{\beta(\epsilon-\mu)}+1)^2} = \frac{N\mathrm{e}^{\beta(\epsilon-\mu)}}{(\mathrm{e}^{\beta(\epsilon-\mu)}+1)^2}$$

[5 points]

Problem 4: Lattice gas model as an Ising model with nonzero H [25 points]

Considering a 2D square lattice with N sites, each site can only occupy a particle and the occupation energy is ϵ. This gives the non-interacting Hamiltonian

$$\mathcal{H}_0 = \sum_{i=1}^{N} \epsilon n_i$$

where $n_i = 0, 1$ is the occupation number for i-th site. In the lattice gas model, the particle in the lattice has attractive interaction with it adjacent particles. This yields the interacting Hamiltonian as

$$\mathcal{H}_{\text{int}} = -\frac{1}{2}\sum_{\langle ij\rangle} u n_i n_j$$

Therefore, the total Hamiltonian is

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{\text{int}} = -\frac{1}{2}\sum_{\langle ij\rangle} u n_i n_j + \sum_{i=1}^{N} \epsilon n_i$$

(a) Express the occupation number n_i in term of spin $\sigma_i = \pm 1$. Rewrite the Hamiltonian $\mathcal{H}$ in term of spin σ_i as

$$\mathcal{H} = -\frac{1}{2}\sum_{\langle ij\rangle} J\sigma_i\sigma_j - H\sum_i \sigma_i$$

what is the relation between J, H and u, ϵ?

Solution

The transformation between σ_i and n_i is

$$n_i = \frac{1+\sigma_i}{2}$$

Therefore, we substitute the above in the the Hamiltonian we have

$$\begin{aligned}
\mathcal{H} &= -\frac{1}{2}\sum_{\langle ij\rangle} u\Big(\frac{1+\sigma_i}{2}\Big)\Big(\frac{1+\sigma_j}{2}\Big) + \sum_{i=1}^{N}\epsilon\Big(\frac{1+\sigma_i}{2}\Big) \\
&= -\frac{1}{2}\sum_{\langle ij\rangle}\frac{u}{4}\sigma_i\sigma_j - \frac{1}{2}\sum_i 2u\sigma_i - \frac{uN}{2} + \sum_{i=1}^{N}\frac{\epsilon}{2}\sigma_i + \frac{N}{2}\epsilon \\
&= -\frac{1}{2}\sum_{\langle ij\rangle}\frac{u}{4}\sigma_i\sigma_j - \sum_{i=1}^{N}\frac{2u-\epsilon}{2}\sigma_i + \frac{N}{2}(\epsilon - u)
\end{aligned}$$

Therefore, we find

$$J = \frac{u}{4}, \quad \text{and} \quad H = \frac{2u-\epsilon}{2} \tag{1}$$

[3 points]

(b) In the mean-field method, we assume the fluctuation of the spin $\delta\sigma_i = \sigma_i - \langle\sigma_i\rangle$ is small where $\langle\sigma_i\rangle = m$ is a average spin on site i. Neglecting $\delta\sigma_i\delta\sigma_j$, we approximate $\mathcal{H} \approx \mathcal{H}_{MF}$ to a non-interacting spin system. Using $\mathcal{H}_{MF}$, calculate the partition function of a canonical ensemble.

Solution

In mean-field approximation, we let

$$\sigma_i\sigma_j = (\delta\sigma_i + \langle\sigma_i\rangle)(\delta\sigma_j + \langle\sigma_j\rangle) = \delta\sigma_i\delta\sigma_j + \langle\sigma_i\rangle\delta\sigma_j + \delta\sigma_i\langle\sigma_j\rangle + \langle\sigma_i\rangle\langle\sigma_j\rangle$$

Using $\delta\sigma_i = \sigma_i - \langle\sigma_i\rangle$, we rewrite the above as follows

$$\sigma_i\sigma_j = \delta\sigma_i\delta\sigma_j + \langle\sigma_i\rangle(\sigma_j - \langle\sigma_j\rangle) + (\sigma_i - \langle\sigma_i\rangle)\langle\sigma_j\rangle + \langle\sigma_i\rangle\langle\sigma_j\rangle$$

This leads to

$$\sigma_i\sigma_j = \delta\sigma_i\delta\sigma_j + \langle\sigma_i\rangle\sigma_j + \sigma_i\langle\sigma_j\rangle - \langle\sigma_i\rangle\langle\sigma_j\rangle$$

Therefore, we rewrite the Hamiltonian

$$\mathcal{H} \approx \mathcal{H}_{MF} = -\frac{1}{2}\sum_{\langle ij\rangle} J\Big(\langle\sigma_i\rangle\sigma_j + \sigma_i\langle\sigma_j\rangle - \langle\sigma_i\rangle\langle\sigma_j\rangle\Big) - \sum_{i=1}^{N} H\sigma_i + N(2J - H)$$

Using $\langle\sigma_i\rangle = m$, we obtain

$$\begin{aligned}\mathcal{H} \approx \mathcal{H}_{MF} &= -\sum_{i=1}^{N} 2J\Big(2m\sigma_i - m^2\Big) - \sum_{i=1}^{N} H\sigma_i + N(2J - H) \\ &= -\sum_{i=1}^{N}(4mJ + H)\sigma_i + N(2J - H + 2Jm^2)\end{aligned}$$

The partition function

$$\begin{aligned}\mathcal{Z} &= \sum_{\sigma_1=\pm1} \cdots \sum_{\sigma_N=\pm1} \exp\Big[\beta\sum_{i=1}^{N}(4mJ + H)\sigma_i - \beta N(2J - H + 2Jm^2)\Big] \\ &= \sum_{\sigma_1=\pm1} \cdots \sum_{\sigma_N=\pm1} \prod_{i=1}^{N} \exp\Big[\beta(4mJ + H)\sigma_i - \beta(2J - H + 2Jm^2)\Big] \\ &= \prod_{i=1}^{N} \sum_{\sigma_i=\pm1} \exp\Big[\beta(4mJ + H)\sigma_i - \beta(2J - H + 2Jm^2)\Big] \\ &= \prod_{i=1}^{N} \mathrm{e}^{-\beta(2J-H+2Jm^2)}(\mathrm{e}^{\beta(4mJ+H)} + \mathrm{e}^{-\beta(4mJ+H)})\end{aligned}$$

We therefore, obtain

$$\mathcal{Z} \approx \Big(\mathrm{e}^{-\beta(2J-H+2Jm^2)}(2\cosh[\beta(4mJ + H)]\Big)^N \quad \textbf{[7 points]}$$

(c) Show that the self-consistent for $\langle \sigma_i \rangle = m$ is

$$m = \tanh\left[4\beta J\Big(m + \frac{H}{4J}\Big)\right]$$

Solution

Following the Ising model calculation in the lecture, we compute

$$\begin{aligned} m =& \frac{1}{\mathcal{Z}} \sum_{\sigma_1=\pm 1} \cdots \sum_{\sigma_N=\pm 1} \sigma_j \exp\left[\beta \sum_{i=1}^{N} (4mJ + H)\sigma_i - \beta N(2J - H + 2Jm^2)\right] \\ =& \frac{1}{\mathcal{Z}} \sum_{\sigma_j=\pm 1} \sigma_j \exp\left[\beta(4mJ + H)\sigma_i - \beta(2J - H + 2Jm^2)\right] \\ & \times \prod_{i=1, i\neq j}^{N} \sum_{\sigma_i=\pm 1} \exp\left[\beta(4mJ + H)\sigma_i - \beta(2J - H + 2Jm^2)\right] \\ =& \frac{e^{-\beta(2J-H+2Jm^2)}\left(e^{\beta(4mJ+H)} - e^{-\beta(4mJ+H)}\right)}{e^{-\beta(2J-H+2Jm^2)}\left(e^{\beta(4mJ+H)} + e^{-\beta(4mJ+H)}\right)} = \tanh[\beta(4mJ + H)] \end{aligned}$$

[5 points]

(d) Find the transition temperature T_C and m near T_C. What is the average occupation number $\langle n_i \rangle$ near T_C?

Solution

$H = 0$, if $2u = \epsilon$. In this case, self-consistent equation is

$$m = \tanh(4\beta m J) \tag{2}$$

Near transition temperature, $m \approx 0$, then we can expand the tanh function as

$$m \approx 4\beta m J - \frac{1}{3}(4\beta m J)^3 \tag{3}$$

We then find

$$m = \sqrt{\frac{3}{(4\beta J)^3}(1 - 4\beta J)} \tag{4}$$

The transition temperature can be obtained by setting the above equation $m = 0$. This leads to

$$1 - 4\beta J = 0 \Rightarrow T = \frac{4J}{k_B} = T_C$$

[3 points]

(e) At $H = 0$, considering m is small near T_C, expand the Helmholtz free energy F up m^4. Express F in term of the mean occupation $n = \langle n_i \rangle$.

Solution

$$F[m] = -k_B T \ln \mathcal{Z} = -Nk_B T \ln \mathrm{e}^{-\beta(2J+4Jm^2)}(2\cosh[\beta(4mJ)]$$
$$= Nk_B T\Big[\beta(2J+4Jm^2) - \ln 2 - \ln\cosh[\beta(4mJ)]$$

Using the Taylor series

$$\cosh(x) = 1 + \frac{1}{2!}x^2 + \frac{1}{4!}x^4 + \dots$$

$$F[m] \approx Nk_B T\Big[\beta(2J+2Jm^2) - \ln 2 - \ln[1 + \frac{1}{2!}(4\beta mJ)^2 + \frac{1}{4!}(4\beta mJ)^4]$$

Using the Taylor series

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \dots$$

$$F[m] \approx Nk_B T\Big[\beta(2J+2Jm^2) - \ln 2 - (\frac{1}{2!}(4\beta mJ)^2 + \frac{1}{4!}(4\beta mJ)^4) + \frac{1}{2}(\frac{1}{2!}(4\beta mJ)^2 + \frac{1}{4!}(4\beta mJ)^4)^2]$$
$$\approx Nk_B T\Big[\beta(2J+2Jm^2) - \ln 2 - \frac{1}{2!}(4\beta mJ)^2 - \frac{1}{4!}(4\beta mJ)^4 + \frac{1}{2}(\frac{1}{2!^2}(4\beta mJ)^4\Big]$$
$$\approx Nk_B T\Big[2\beta J - \ln 2 + (2\beta J - \frac{1}{2!}16\beta^2 J^2)m^2 + [-\frac{4^4}{24} + \frac{4^4}{2\times 2^2}]m^4(\beta J)^4\Big]$$
$$\approx Nk_B T\Big[2\beta J - \ln 2 + 2\beta J(1-4\beta J)m^2 + \frac{1}{12}m^4(4\beta J)^4\Big]$$

We can write $4\beta J = \frac{T_C}{T}$ where $T_C = 4J/k_B$. This gives

$$F[m] \approx Nk_B T\Big[2\beta J - \ln 2 + \frac{1}{2}(T/T_c)(1-T/T_C)m^2 + \frac{1}{12}m^4(T/T_C)^4\Big]$$

Substitute $m = 2n-1$, we obtain

$$F[m] \approx Nk_B T\Big[2\beta J - \ln 2 + \frac{1}{2}(T/T_c)(1-T/T_C)(2n-1)^2 + \frac{1}{12}(2n-1)^4(T/T_C)^4\Big]$$

[7 points]